

BenthicComplexVision (BCV) — User Guide

BenthicComplexVision (BCV) is an open-access, AI-based web tool that estimates the **structural complexity** of benthic habitats from a **single top-down image**. In a few seconds it produces three key metrics — **Gradient Rugosity**, **Fractal Dimension**, and **Colony Height** — together with an interactive 3D reconstruction of the surface.

This guide walks you through the complete workflow, from opening the page to exporting your results.

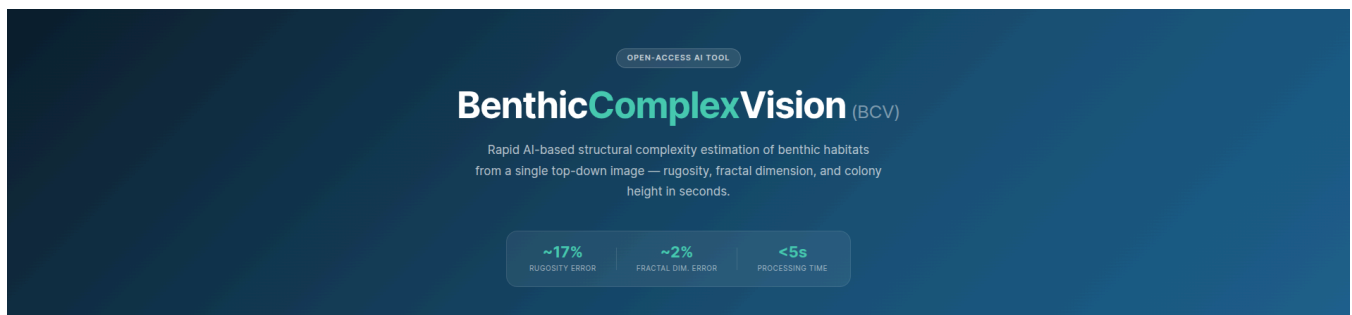


Table of Contents

1. [Overview of the Workflow](#)
2. [Step 1 — Add an Image](#)
3. [Step 2 — Navigate and Zoom](#)
4. [Step 3 — Set the Real-World Scale \(Reference Points\)](#)
5. [Step 4 — Draw a Bounding Box](#)
6. [Step 5 — Run the Estimate](#)
7. [Step 6 — Read the Complexity Metrics](#)
8. [Step 7 — Explore the 3D Viewer](#)
9. [Exporting Results](#)
10. [Saving and Loading Projects](#)
11. [Working with Multiple Images](#)
12. [Troubleshooting & Tips](#)

1. Overview of the Workflow

The tool guides you through a simple, repeatable sequence:

1. Add Image
2. Set Area of Interest
3. Set Reference Point (Optional)
4. Estimate Structural Complexity
5. Export / Save Project

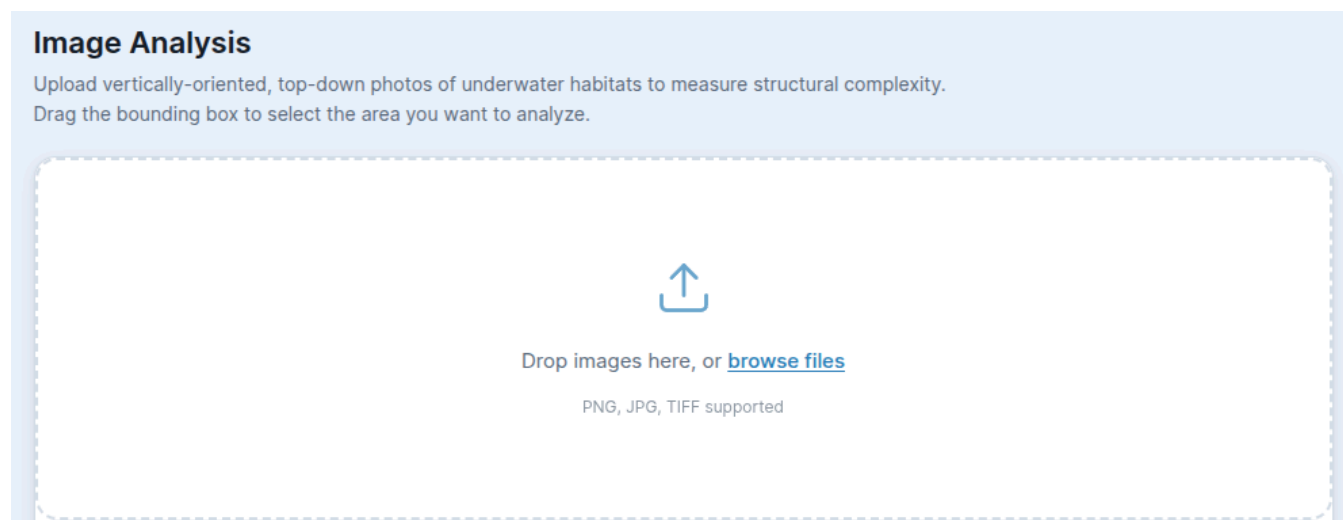
You can analyse one image or batch several images in a single project, switching between them at any time. Each image keeps its own reference points, bounding box, depth map, and results.

2. Step 1 — Add an Image

When you first open BCV, the image panel is empty and offers three ways to get started.

Option A — Upload your own image

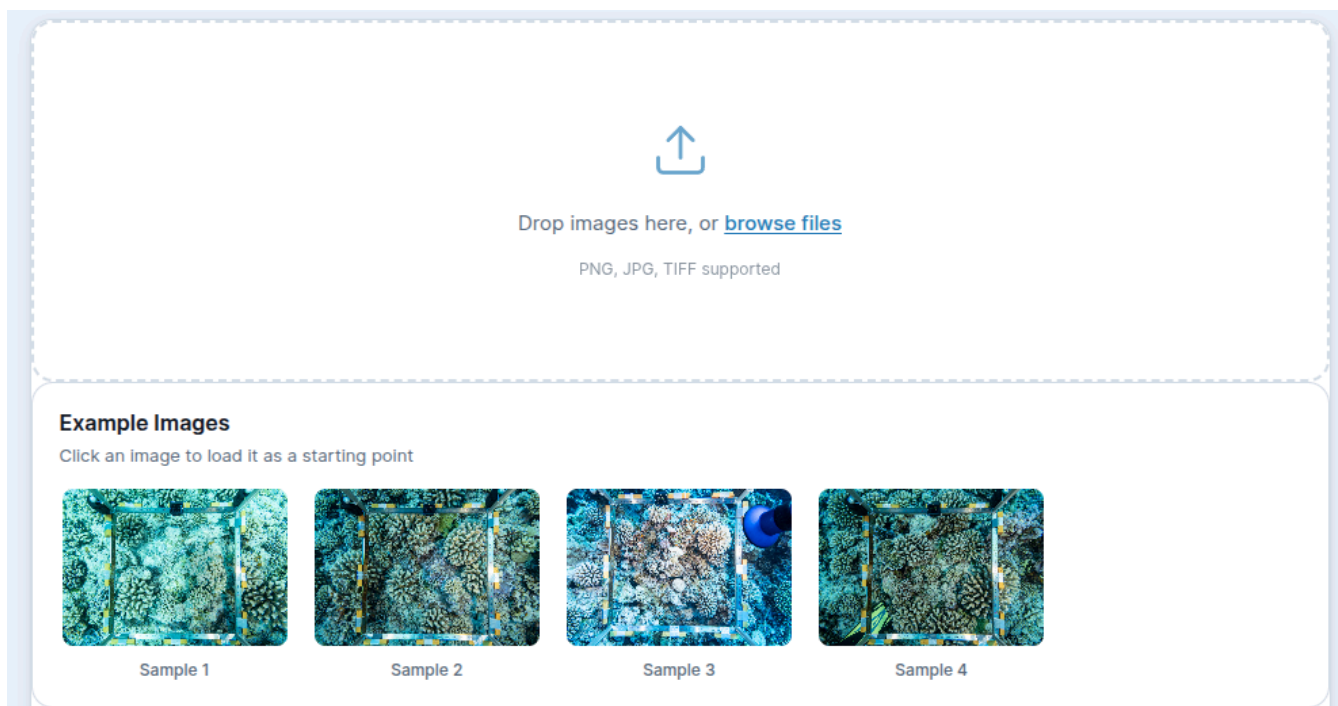
- **Drag and drop** one or more image files onto the drop area, **or**
- Click "**browse files**" to open your file picker.
- Supported formats: **PNG, JPG, TIFF**.



 **Tip:** For the most accurate results, use a clear, well-lit, **top-down** image where the seabed/colony fills most of the frame and the camera is roughly parallel to the surface.

Option B — Use a built-in example for quick demonstration

If you just want to try the tool, load one of the four bundled sample images. Each sample has a "**Load**" button that adds it straight into your project.



Option C — Open a saved project

If you have a previously saved `.mbct` project file, click **"Load existing project (.mbct)"** (or drag the `.mbct` file onto the drop area) to restore all of its images, annotations, and results. See [Saving and Loading Projects](#).

3. Step 2 — Navigate and Zoom

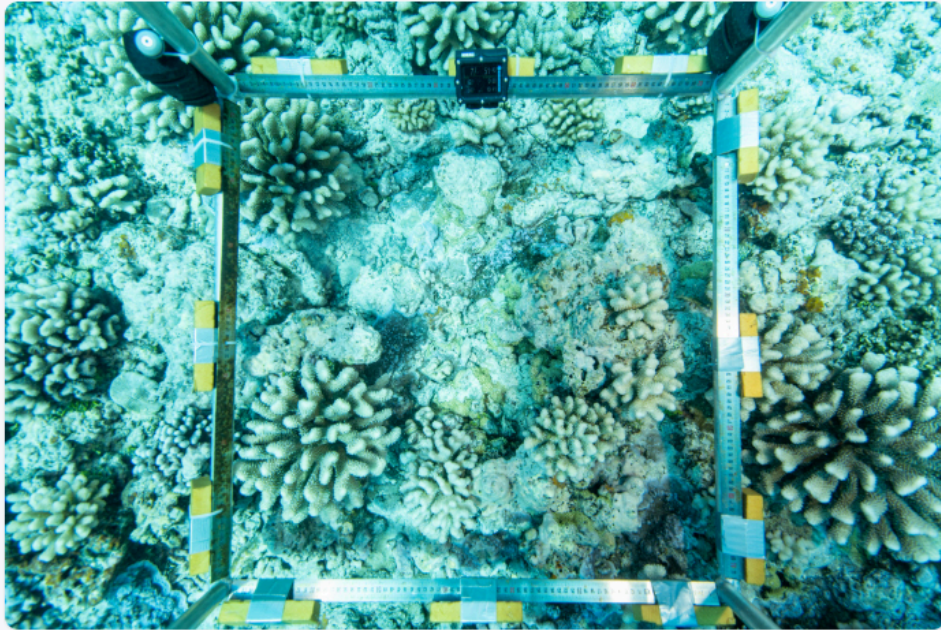
Once an image is loaded, it appears in the large central viewer with a thumbnail strip below.

Action	How
Zoom in / out	Scroll the mouse wheel over the image (5%–100% of full size)
Pan	Hold the middle / right mouse button and drag
Reset zoom to fit	Double-click the image
Switch images	Click a thumbnail, or use the on-image ◀/▶ arrows
Add more images	Click the + ("Add more") tile in the thumbnail strip
Remove current image	Click the ✕ button near the image info

The image info area shows the current filename and position in the set (for example, `1 / 4`).

Image Analysis

Upload vertically-oriented, top-down photos of underwater habitats to measure structural complexity.
Drag the bounding box to select the area you want to analyze.



sample_1.jpg 1/1 ×



SAVED BOXES

Full Image
2048 × 1366

4. Step 3 — Draw a Bounding Box

The **bounding box** defines the rectangular region of the image that will be analysed. A bounding box is **required** before you can run an estimate.

Draw a box

- **Left-click and drag** on the image to draw a rectangle. Release to finalise it.
- The minimum size is a few pixels — make the box large enough to enclose the structure you care about.

Adjust a box

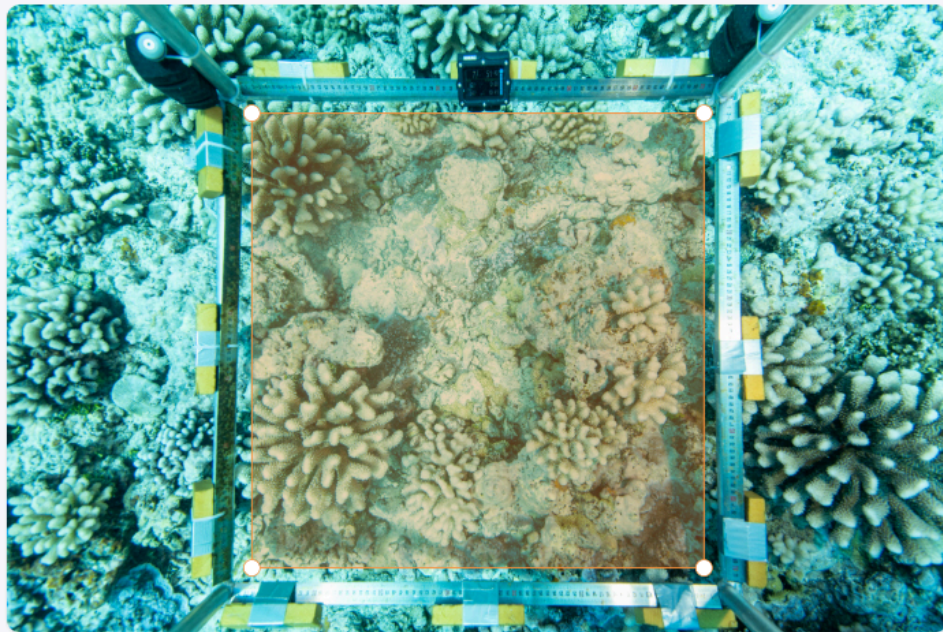
- **Move:** click and drag the centre of the box.
- **Resize:** drag any of the four corner handles.

Reuse boxes (Saved Boxes)

- Every box you draw is automatically stored in the **Saved Boxes** list, which shows a preview and the box's pixel dimensions.
- Click a saved box to re-apply it (centred) to the current image — handy for keeping a consistent analysis area across multiple images.
- Use **"Full Image"** to set the bounding box to the entire image.
- Remove a saved template with its ✕ button.

Image Analysis

Upload vertically-oriented, top-down photos of underwater habitats to measure structural complexity.
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sample_1.jpg 1/1 ✕



+

SAVED BOXES

Full Image
2048 × 1366

983 × 989
pixels

5. Step 4 — Set the Real-World Scale (Optional)

Because a photo has no inherent size, BCV needs at least **two reference points** with a known real-world distance between them to calibrate scale. This is what makes **Colony Height** (and the absolute geometry of the 3D model) meaningful.

Place reference points

1. Click **"Add Point"** (the target icon). The button switches to **"Stop Adding"** while you are in point-placement mode.
2. Click on the image to drop a point. Points appear as **green circles**.
3. Place **at least two** points at locations whose real-world separation you know (for example, the two ends of a ruler or scale bar laid in the scene).

Enter the distance for a point

1. Click a placed point — it turns **blue** to show it is selected and a small popup opens.
2. In the popup (labelled **"Point 1"**, **"Point 2"**, ...), type the **distance** value.
3. Choose the **unit** from the dropdown: **cm**, **mm**, or **m**.
4. Press **Enter** (or click away) to save. Use the popup's delete control to remove a single point.

Manage points

- **"Clear Points"** removes all reference points on the current image. (Disabled when there are none.)
- Click **"Stop Adding"** when you are finished placing points so that normal click/drag returns to bounding-box editing.

Image Analysis

Upload vertically-oriented, top-down photos of underwater habitats to measure structural complexity.
Drag the bounding box to select the area you want to analyze.



SAVED BOXES

Full Image
2048 × 1366

983 × 989
pixels

Load Project

Clear Points

Stop Adding

Estimate

⚠ Important: If fewer than two reference points are set, BCV still computes rugosity and fractal dimension (which are scale-independent), but scaled outputs such as colony height depend on having a valid two-point scale.

6. Step 5 — Run the Estimate

When you have (1) an image, (2) at least two reference points with distances, and (3) a bounding box, click **"Estimate"** (the bar-chart icon).

BCV then performs two operations automatically:

1. **Depth prediction** — an AI depth model reconstructs the 3D shape of the cropped region.
2. **Complexity analysis** — the metrics are computed from that depth map and your reference scale.

While it runs, the 3D Viewer shows a spinner and the label **"Estimating..."** (this typically takes only a few seconds).

[Load Project](#)[Clear Points](#)[Stop Adding](#)[Estimate](#)

 **Re-running:** If you only change the reference points, BCV re-computes the metrics using the existing depth map (fast). If you change the image, bounding box, or depth model, it runs a full new depth prediction.

7. Step 6 — Read the Complexity Metrics

After estimation, the **Complexity Metrics** table (the Analysis Report) appears with three rows:

Metric	What it means	Format
Gradient Rugosity	Ratio of surface area to planar area (higher = rougher)	3 decimals
Fractal Dimension	3D structural complexity measure	3 decimals
Colony Height	Maximum vertical extent	2 decimals, shown in your chosen unit (cm/mm/m)

Complexity Metrics			Export	Export (All Image)
METRIC	VALUE	DESCRIPTION		
Gradient Rugosity	1.540	Ratio of surface area to planar area		
Fractal Dimension	2.223	3D structural complexity measure		
Colony Height	12.48 cm	Maximum vertical extent		

8. Step 7 — Explore the 3D Viewer

The **3D Viewer** shows an interactive reconstruction of the analysed region. Use the toggle buttons at the top to switch between three modes:

- **Point Cloud** — the 3D points, textured with the original image colours.
- **Mesh** — a shaded, triangulated surface.
- **Depth Map** — a 2D heatmap of depth (red = closest, blue = farthest).

Your reference points appear as **green spheres** in every mode so you can see where they sit on the surface.

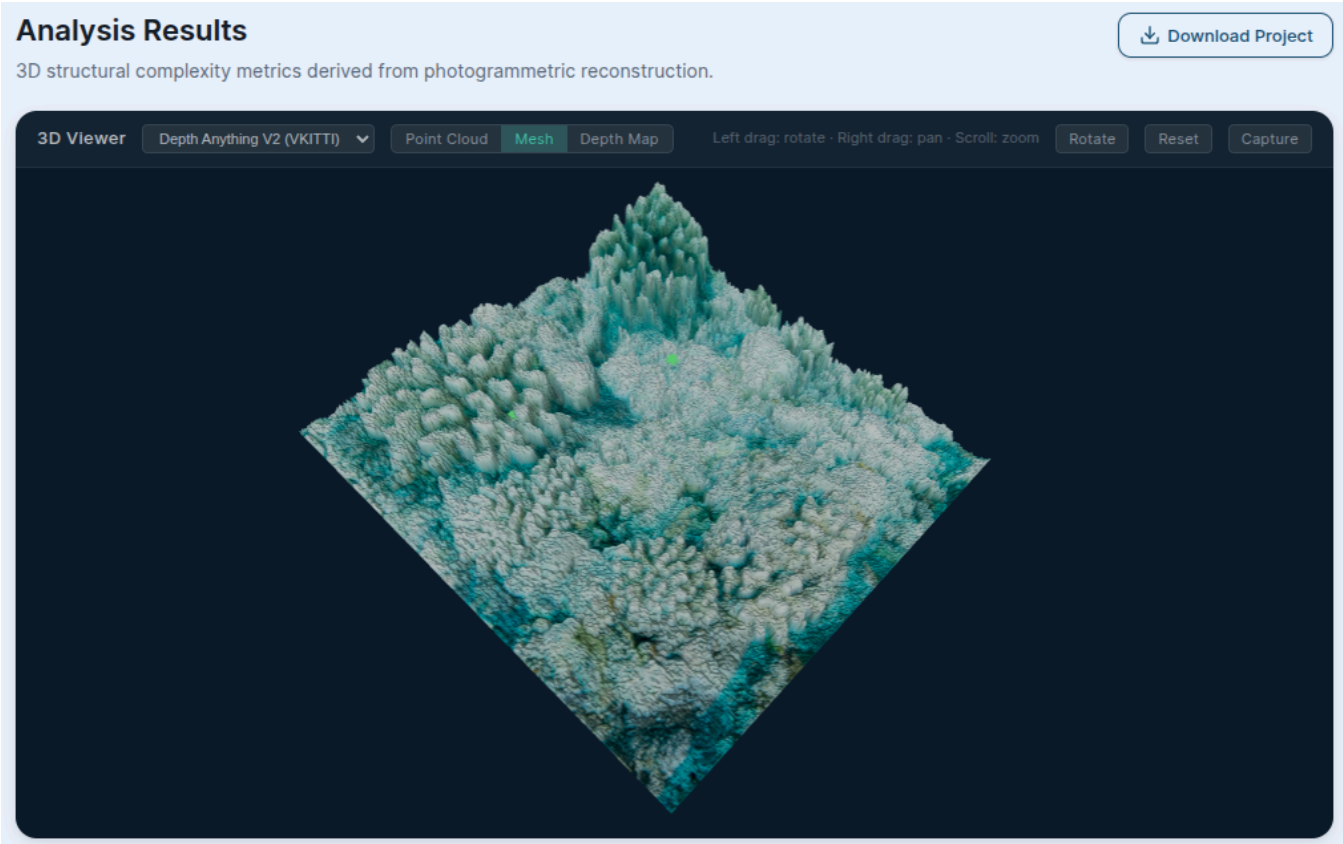
Camera controls

The hint bar reads: "**Left drag: rotate · Right drag: pan · Scroll: zoom**".

Control	Action
Left-click drag	Rotate the model
Right-click / middle drag	Pan the view
Scroll wheel	Zoom in / out

Viewer buttons

- **Model dropdown** — switch the AI depth model between "**Depth Anything V2**" and "**Depth Anything V2 (VKITTI)**". Changing the model re-runs the estimate.
- **Rotate** — toggles continuous auto-rotation on/off (pauses while you drag).
- **Reset** — returns the camera to its default position (does not recompute anything).
- **Capture** — saves a high-resolution PNG screenshot of the current view (filename includes the view mode and a timestamp, e.g. `screenshot-mesh-<timestamp>.png`).



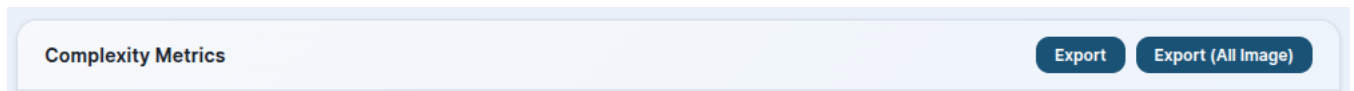
9. Exporting Results

In the **Complexity Metrics** header you have two CSV export options:

- **"Export"** — downloads the current image's metrics as `analysis_report_<imagename>.csv`.

- **"Export (All Image)"** — downloads metrics for every image in the project as `analysis_report_all_images.csv`.

Each CSV row includes the image name, the bounding box (x, y, width, height), the analysed area, and the three metrics (rugosity, fractal dimension, colony height).



10. Saving and Loading Projects

BCV can save your entire session — including images, reference points, bounding boxes, depth maps, and computed metrics — to a single project file.

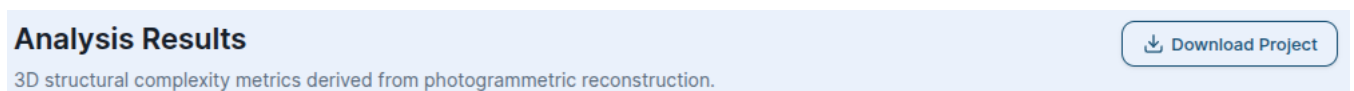
Save

Click **"Download Project"** (download icon) to save everything as a `.mbct` file. You can rename the file after download.

Load

- From the empty start screen: click **"Load existing project (.mbct)"** (or drag the `.mbct` file onto the drop area).
- With images already loaded: click **"Load Project"**.

When a project is loaded, BCV restores all images and results and switches to the first image.



11. Working with Multiple Images

You can analyse many images in one project:

1. Add several images (drag a batch in, or use the + "Add more" tile).
2. Select each image from the thumbnail strip and give it its own reference points and bounding box. The **Saved Boxes** list makes it easy to apply a consistent box to each image.
3. Run **Estimate** on each image. Results are stored per image.
4. Use **"Export (All Image)"** to get a single CSV containing every image's metrics.
5. Save the whole set with **"Download Project"**.

12. Troubleshooting & Tips

Situation	What it means / what to do
"Bounding box required" message	You clicked Estimate without drawing a bounding box. Draw one on the current image first.

Situation	What it means / what to do
"Estimation failed"	A server-side error occurred during depth prediction or analysis. Retry; check your connection or try a different depth model.
Colony Height looks wrong	Verify you placed two reference points and entered the correct distance and unit .
Image won't upload	Confirm the file is PNG, JPG, or TIFF .
Model looks too flat / too exaggerated	Try switching the depth model in the 3D Viewer dropdown and re-estimating.

Best-practice checklist

-  Use a **top-down**, sharp, evenly lit image.
-  Include a **physical scale** (ruler / scale bar) so reference points are accurate.
-  Place reference points precisely on the known-distance markers.
-  Keep the bounding box tight around the structure of interest.
-  Save your work as a `.mbct` project so you can revisit or refine it later.